# LPUtility - A electron web-based application for Computer Studies Students that provides All-In-One Software and Utilities Commonly used in Academics and IT

**Marc Lorenz Alvarez, Jose Karlo Fernandez, Christofferson Kyle C. Garcia, Raphael Herrera, Hench Lloyd Marquez, Joseph Pimentel, Josiah Owen Pinca, and Yuan Aundre Rubio**
College of Information Technology and Computer Studies, Lyceum of the Philippines University Cavite

**Keywords:** API, Scheduler, Utility, Web-Based, CLI, Productivity, Technology, Computer Studies

## I. Introduction

LPUtility is a web application for Information Technology and Computer Science students. It puts the software and utilities that IT courses commonly ask for into one responsive website, so a student does not have to jump between many sites and installers. It is built with HTML, CSS, JavaScript, and ReactJS, which means it runs in a normal browser on a laptop or a phone.

The proposed parts are a software catalog with a Chocolatey command builder, a schedule maker, a password generator, an IPv4 generator, and an IPv6 calculator. The catalog covers programming software, IT tools, and academic tools. The last three parts are for networking and cybersecurity practice.

Two definitions before going further. A package manager installs software from a typed command, instead of the user visiting a website and clicking through an installer. Chocolatey is a package manager for Windows, and Winget is Microsoft's own (Bisson, 2021). Both run on the student's computer, and a normal website cannot install programs on someone's PC. So LPUtility prepares the command and the student runs it. The rest of this paper is written with that limit in mind.

## II. Problem Background

Salerno et al. (2025) observed 18 students installing seven common development tools across 24 sessions, then surveyed 144 more students to check the results. In the survey, 88 students said they had run into unclear installation instructions and 62 said the instructions did not work. The most common problem in the sessions was a complicated installation process, which showed up in 14 of the 24 sessions. The authors also point to earlier studies where participants said they sometimes spent hours or even days installing and configuring tools. In plain words, the problem is lost time and lost motivation before any real coding happens.

Linux users have long had package managers, while Windows went a long time without a built-in one (Bisson, 2021). Chocolatey and winget now cover that gap, but both are command line tools. As we see it, a first-year student still has to know the package names and the syntax, and has to trust whatever

tutorial they find. The gap LPUtility targets is the one between "a package manager exists" and "a beginner can use it on the first day."

APNIC reported that IPv6 capability in the Philippines was 17.66% in April 2025, while the APNIC region as a whole passed 50% (Low, 2025). "Capable" here only means a test could fetch a page over IPv6, so it is not the same as people using it every day. Still, it suggests a Filipino student may meet IPv6 mostly inside the classroom. On the security side, ISC2 (2025) surveyed 16,029 cybersecurity professionals and found that 95% had at least one skill need and 59% had critical or significant ones.

LPUtility will not fix a global skills shortage. The realistic claim is smaller: subnetting, IPv6 notation, and password strength are basics that can be practiced for free with instant answers. A student project from ESPOL in Ecuador built a React-based subnetting game because students find the topic hard (Barreto Anzules & Ponce Clavijo, 2023). It is an undergraduate project with its own survey, so we treat it as an example of the same idea and not as proof that it works.

Liu et al. (2026) combined 31 studies with 13,506 college students and found a moderate positive relationship between time management and learning outcomes ($r = .25$), strongest among undergraduates. It is a correlation, and the results vary a lot from study to study, so a schedule maker will not raise anyone's grades by itself. What it can do is make planning easier to start.

Jaca (2022) studied Filipino state university students during remote learning and reported poor, unstable internet and limited money for mobile data as major problems. The sample was small (60 education students in Cebu) and the setting was the pandemic, so we only used it as a warning. LPUtility should be light, should use an account, and should keep working on a weak connection once loaded. The catalog only produces text, so the site itself stays small, but the software the student then downloads still uses their data.

## III. Significance of the Project

LPUtility is useful if it cuts the time between "I have to install these tools" and "I can start the lab," and if it puts small practice tools in the same place. It is not useful for everyone, so Table 1 shows who it is for and Table 2 shows who it is not for.

**Table 1**
*Intended Beneficiaries*

| Who | Why they benefit | Basis |
|---|---|---|
| Computers studies students on a personal Windows laptop | They must install several tools for different courses and have little experience with installers or the terminal. | Salerno et al. (2025) |

| Who | Why they benefit | Basis |
|---|---|---|
| Networking and cybersecurity students | They need repeated practice in subnetting, IPv6 notation, and password strength, with instant feedback. | Low (2025); ISC2 (2025); Barreto Anzules & Ponce Clavijo (2023) |
| Students with a busy or irregular timetable | They need a quick way to lay out class and study blocks and see clashes. | Liu et al. (2026) |
| Instructors (indirect) | They could share one ready software list instead of a long setup handout. This is our proposal only and it is untested. | Proposal, no source |

**Table 2**
*Groups Who Will Benefit Little*

| Group | Reason |
|---|---|
| Windows, macOS and Linux users | The Chocolatey command builder is for Windows. |
| Experienced users | They already script installs with choco or winget. |
| School computer labs managing many PCs | Organizational use of the Chocolatey Community Repository has its own rules (Valdinger, 2026). |

If we narrow it down to one person, LPUtility is for a first- or second-year IT student with their own Windows laptop, limited mobile data, and nobody nearby to ask, who has to install several tools before the first lab. At its deepest, the problem is not that the software is hard to find. The steps are scattered, unclear, and easy to get wrong, and a beginner has no easy way to tell whether the install worked. Salerno et al. (2025) list unclear instructions and missing installation feedback among the main challenges.

## IV. Target Users

**Table 3**
*Users, Needs, and Where the Application Can Fail*

| User | What they need | What LPUtility gives | Where it can fail |
|---|---|---|---|
| IT/CS student | A short list of trusted tools, commands that work, and a way to check the install. | Catalog with bundles, one copy-ready command, and verify commands. | Needs an administrator terminal. A school-managed laptop may block installs. Many packages at once can hit the Chocolatey rate limit (King, 2026). |
| Networking or cybersecurity student | Practice problems with correct answers and quick conversions. | IPv4 generator with worked answers, IPv6 calculator, password generator. | A wrong calculator teaches wrong answers, so testing matters. |
| Student planning a semester | A weekly timetable that warns about clashes and can be saved or printed. | Schedule maker with local save and export. | Browser data can be cleared. No reminders in version 1. |
| Instructor (secondary user) | A ready list of software to give a class. | A copyable software list. | Untested. Depends on the catalog staying current. |

## V. Proposed Solution

The proposed solution is one static website that turns scattered setup work into steps a beginner can follow. It rests on four decisions.

**1. Generate, do not install.** The site never touches the student's PC. It builds a `choco install` line from the tools the student ticked, plus a short list of verified commands, for example `git --version`. Salerno et al. (2025) recommend giving users a test to confirm the install worked and telling them up front when several tools will be installed. We follow both.

**2. Selected, not everything.** The catalog stays small, with every entry checked, and is split into programming software, IT tools, and academic tools. A short list is easier to keep correct and easier for a beginner to trust.

**3. Run in the browser and app.** The tools do their work in the student's electron based application. Saved work stays in that app. This keeps the app light for students with weak connections (Jaca, 2022).

**4. Say the limits out loud.** Each tool shows what it does not do: Windows only, administrator terminal needed, rate limits apply, practice addresses are not real assignments.

To find out if it works, we would run a small pilot with one class, compare the time to finish a setup with and without the catalog, and count failed installs. Nothing like this has been done yet.

## VI. Core Features

There are five proposed features. Each one says what it does and what it cannot do.

### 1. Software Catalog and Chocolatey Command Builder

**What it does.** The student browses programming software, IT tools, and academic tools, ticks what they need, and gets one copy-ready Chocolatey command and a list of verified commands. Ready-made bundles, such as a first-year programming set, mean the student does not start from zero. The catalog is a JSON file in the project. Each Chocolatey package ID is checked with the Chocolatey CLI on a schedule, which is one of the two supported ways to query the repository (Broadwith, 2024). The website itself does not call Chocolatey while a student is using it.

**Limits.** Windows only. The student needs Chocolatey installed and usually an administrator terminal. Chocolatey limits package downloads to 20 per minute per IP address, and goes over blocks that IP for one hour (King, 2026). Students on a shared school network could hit this when many people install at once. That is our own reading of the rule and not something the sources tested. Bundles will stay small for this reason.

### 2. Schedule Maker

**What it does.** The student enters subjects with day, time, room, and color, and gets a weekly grid. The app flags overlapping times, lets the student print the grid or save it as an image, and exports an .ics file that calendar apps can import. Data is saved per account.

**Limits.** The schedule is created through image to text. The resolution of the image can cause the image to not always be understood.

### 3. Password Generator

**What it does.** It creates random passwords in the browser using the Web Crypto API instead of Math.random(). The student picks the length and character sets, and the app shows a simple strength estimate (length times the base-2 log of the character pool size). NIST's current guidance says a password used alone must be at least 15 characters, allows shorter ones only as part of multi-factor authentication, and says systems must not force composition rules such as "must include a symbol" (Temoshok et al., 2025). So the default length is 16 and the app warns below 15. Nothing is sent or stored.

**Limits.** NIST's rules are for systems that check passwords, not for generators, so we only borrow the length advice. A generator does not stop password reuse or phishing, and some websites still cap password length.

**4. IPv4 Generator**

**What it does.** It generates random valid IPv4 addresses with a CIDR prefix for practice, using the private ranges by default. For each one it shows the subnet mask, network address, broadcast address, first and last usable host, and number of usable hosts. A practice mode gives the problem first and reveals the answer after the student tries. It can also split one block into several subnets.

**Limits.** The addresses are for practice and are not assigned to anyone. Real public addresses come from registries such as APNIC (Low, 2025). The rules behind the private ranges are older standards from before 2020, so they are left out of the reference list on purpose.

**5. IPv6 Calculator**

**What it does.** The student types an IPv6 address with a prefix length. The app shows the full and shortened forms, the network prefix, the first and last address of the prefix, how many /64 subnets fit inside a shorter prefix, and the address type (loopback, link-local, unique local, multicast, or global unicast). It uses JavaScript BigInt because IPv6 addresses are 128 bits long.

**Limits.** It only calculates. It does not test whether an address is reachable or assigned. Since only 17.66% of Philippine users were IPv6 capable in April 2025 (Low, 2025), many students may have no IPv6 network to try things on, so worked examples inside the app matter. The address formatting rules also come from older standards, so they are not in the reference list.

## VII. Authentication Requirement

LPUtility depends on one outside service, the Chocolatey Community Repository, and needs no OAuth in version 1.

**Table 4**
*External Services and Access Decisions*

| Item | What we found | What we do |
|---|---|---|
| Repository access | Only Chocolatey CLI and chocolatey.lib are supported ways to query it. Custom queries may break, results are capped at 10,000 items, and unsupported filters return an HTTP 406 error (Broadwith, 2024). | The website makes no live calls. A scheduled check with the Chocolatey CLI validates the package IDs in our catalog. |
| Rate limits | 20 package downloads per minute per IP address, and 5 per minute for downloading Chocolatey CLI itself. Going over causes a one-hour block, and triggering it again restarts the timer (King, 2026). | Keep bundles small and show the limit on the page. |
| API key | We found no API key requirement in the Chocolatey material we read. The repository is described as a public resource (King, 2026). | Not needed for Chocolatey, since we do not call the API at runtime. |
| OAuth sign-in | RFC 9700 (Lodderstedt et al., 2025) is current security guidance for OAuth 2.0 and discourages some older modes, including the implicit grant. | Optional sign-in with third-party providers (Google and GitHub are the planned options) using the authorization code flow with PKCE. It goes through a managed authentication service so no client secret sits in the React code. LPUtility never sees the user's provider password. |
| Guest access | This is our own design decision, based on the connection and data limits in Jaca (2022). | "Continue as guest" opens every tool with no sign-in. Guest data stays in the browser (localStorage). A guest who signs in later can upload the browser data to the new account. |
| Data kept for signed-in users | This is our own design decision. | Only the name and email from the provider, the saved schedule, and saved software lists. Generated passwords and practice problems are never uploaded. |
| Public access | The site is public and free, and every tool works in guest mode. | LPUtility has no public API of its own in version 1. Only the sign-in and saved-data service sits behind an account. |

## VIII. Proposed Technologies, Languages, Frameworks, and Tools

**Table 5**
*Proposed Technology Stack*

| Layer | Choice | Reason |
|---|---|---|
| Markup and styling | HTML5 and CSS3 (flexbox, grid, media queries) | Required by the project. Responsive layout for laptop and phone. |
| Language | JavaScript (ES2020 or newer) | Required by the project. Native BigInt handles 128-bit IPv6 math. |
| Framework | ElectronJS with function components and hooks. | Reusable UI parts for each tool. The ESPOL subnetting project also used Electron (Barreto Anzules & Ponce Clavijo, 2023). |
| Storage | Electron state and browser localStorage | Saves schedules without a backend or database. |
| Data | JSON software catalog: name, category, Chocolatey package ID, verify command | Easy to review and fix in the repository. |
| Randomness | Web Crypto API (`crypto.getRandomValues`) | Cryptographically secure random values for passwords. |
| Package validation | Chocolatey CLI run on a Windows machine on a schedule | A supported way to query the repository (Broadwith, 2024). |
| Testing | Vitest and React Testing Library | Known subnetting and IPv6 answers can be checked automatically. |
| Version control and hosting | Git and GitHub, with a static host such as GitHub Pages | Static files only. No server to maintain. |
| Optional | Service worker (PWA) for offline use | Helps students with weak connections (Jaca, 2022). Not required for version 1. |

**IX. References**

Barreto Anzules, A. E., & Ponce Clavijo, B. S. (2023). *Desarrollo e implementación de una herramienta gamming innovadora para potenciar el aprendizaje de subnetting en la materia de redes de datos* [Development and implementation of an innovative gaming tool to strengthen subnetting learning in the data networks course] [Integrative project, Escuela Superior Politécnica del Litoral]. ESPOL Institutional Repository. http://www.dspace.espol.edu.ec/handle/123456789/58169

Bisson, S. (2021, March 17). Getting started with winget, the Windows Package Manager. *InfoWorld*. https://www.infoworld.com/article/2263233

Broadwith, P. (2024, March 15). *Upcoming changes to the Chocolatey Community Repository*. Chocolatey Software Blog. https://blog.chocolatey.org/2024/03/ccr-api-changes/

ISC2. (2025, December 4). *ISC2 publishes 2025 cybersecurity workforce study* [Press release]. https://www.isc2.org/insights/2025/12/isc2-publishes-2025-cybersecurity-workforce-study

Jaca, C. A. L. (2022). State university students' learning locations and remote learning challenges during the COVID-19 pandemic. *International Journal of Learning, Teaching and Educational Research, 21*(2), 195–210. https://doi.org/10.26803/ijlter.21.2.11

King, J. (2026, July 15). *I got 429 problems, but a cache ain't one*. Chocolatey Software Blog. https://blog.chocolatey.org/2026/07/ccr-rate-limiting-explained/

Liu, B., Ma, P., & Jia, F. (2026). Systematic review and meta-analysis of the impact of time management on college students' learning outcomes. *Frontiers in Psychology, 17*, Article 1700298. https://doi.org/10.3389/fpsyg.2026.1700298

Lodderstedt, T., Bradley, J., Labunets, A., & Fett, D. (2025). *Best current practice for OAuth 2.0 security* (RFC 9700). Internet Engineering Task Force. https://www.rfc-editor.org/rfc/rfc9700.html

Low, J. R. (2025, April 23). *IPv6 capability reaches 50% in the Asia Pacific region*. APNIC Blog. https://blog.apnic.net/2025/04/23/ipv6-capability-reaches-50-in-the-asia-pacific-region/

Salerno, L., Treude, C., & Thongtanunam, P. (2025). *Open source software development tool installation: Challenges and strategies for novice developers* (arXiv:2404.14637v3). arXiv. https://arxiv.org/pdf/2404.14637

Temoshok, D., Choong, Y., Regenscheid, A., Galluzzo, R., Fenton, J., Richer, J., & Lefkovitz, N. (2025). *Digital identity guidelines: Authentication and authenticator management* (NIST Special Publication 800-63B-4). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-63B-4

Valdinger, S. (2026, June 1). *Clarifying organizational use of the Chocolatey Community Repository*. Chocolatey Software Blog. https://blog.chocolatey.org/2026/06/updated-terms-of-use/